

First, reproduce learning in a simpler setting.

A recursive language model (RLM) can use code and ask helper models to solve parts of a problem. We eventually want to train those choices.

Today's question

Can we use a published training recipe to improve a model's math answers on questions excluded from training?

We use the authors' code for **Dr. GRPO**, a reinforcement learning (RL) method. It learns from scores assigned to the model's own answers.

This is a small reproduction of an existing method. It establishes a reference for future work on recursive models.

Liu et al., *Understanding R1-Zero-Like Training: A Critical Perspective* (2025).

Official code; our run is much smaller than the paper's experiments.

Reward correct attempts, then change the model.

Illustrative question: Solve $3x + 2 = 14$. The correct answer is $x = 4$.

The model's attempts	Checker score	Learning direction
Two answers end in $x = 4$.	1 each	Make these more likely.
Two end in $x = 5$ or $x = 6$.	0 each	Make these less likely.

Each attempt is compared with the average score for that question. These differences guide changes to the model's numerical settings, or **weights**.

The checker judges the final answer. We do not supply worked solutions for the model to copy.

Simplified illustration with four attempts; our runs use eight. If every attempt receives the same score, this question supplies no relative learning signal.

We train on math, then test on different questions.

We trained **Qwen2.5-Math-1.5B** on 512 math questions, with eight attempts per question and 32 weight updates, on one A100 GPU.

We tested the starting and trained models on the same **500 MATH500 questions**, all excluded from weight training.

We also tested two ways of asking each question.

Chat style: Put the question inside a conversation, with an instruction to explain the steps and mark the final answer.

Question alone: Supply only the original question, with no added conversation or instruction.

Training used chat style; testing used both formats.

Of the 500 questions, 64 were used to monitor progress. We did not choose the best checkpoint. Earlier pretraining exposure is unknown.

The starting prompt changes the size of the gain.

Correct answers out of the same 500 MATH500 questions.

How we ask	Before RL	After RL	Change
Chat style	154	308	+154
Question alone	305	317	+12

Without any new training, giving only the question nearly matched the trained model's chat-style score.

Lesson: RL improved the score, but the large gain does not by itself show that the model learned new mathematical abilities.

One training run; the same final checkpoint, checker, and length limit in both formats. The checker judges final answers, not every reasoning step.

A higher score does not always mean a new skill.

An actual test question: Write $3/20$ as a decimal.

The saved responses, summarized

Before: The model repeated the question until it hit the length limit.

After: It explained the division and answered 0.15.

This came from the earlier 128-question chat-style test. Many gains were recoveries from repetitive responses, not necessarily new mathematical knowledge.

Now running: Start again from the original model and train with the question alone. Can RL improve on that stronger starting point?

For future RLM work: compare against a strong starting setup and inspect what changes. Better math answers do not yet establish better problem decomposition.